A memoryless two-header tape machine: what a window costs, and why two lower-bound techniques fail

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Abstract

A tape of n cells over three values — equivalently, two-bit cells read through their sum, the encoding Section 2 fixes — is read by two headers that enclose a window; at each step a fixed rule, given the window's content and the geometry, chooses one of eight actions — move a header inward or outward, trim a cell from the left, the right or both, or halt. The cost is the number of cell reads, the state is $O(\log n)$ bits, and the question is how the cost must grow with n for each rule. On the evidence the rules fall into three classes — fully linear, semi-linear and superlinear — separated by a worst-case cost and a mean cost. All three are populated: four natural rules are fully linear, one construction is semi-linear ($e = \Theta(n)$ with $g = \Theta(n^2)$), and the remaining seventeen are superlinear, ordered by a measured predictability constant κ_ρ (Section 3). The superlinear class is not split further.

We separate the length of a walk from the number of cells it reads. The length turns out to be a property of the rule rather than of the model: at $n = 4096$ the measured step count ranges from 0.385 to 15.58 times $n^2/128$, a factor of forty, and the growth exponent between $n = 1024$ and 4096 ranges from 1.70 to 2.20. Every exponent quoted here is therefore either fitted from the measured reads at four sizes or derived from a step count where the machine's steps are known. If the rule's output is a group sum of one-cell terms, a three-word machine answers every query in $n + S - 1$ reads, where S is that rule's own step count; six constructions that once looked super-linear are consequences of this one statement, not of any rule.

For rules whose output is a kernel correlation, the reads per query are 1 when the kernel's Hankel rank is at most the number s of linear functionals the state holds, and m/s when it exceeds s; such rules cost $O(n^2)$ when the rank fits the state's functionals and $O(n^3/\log n)$ when it is full, with a machine attaining each; whether the second is optimal is open. For five level-count rules we give machines measured at n up to 16384 whose reads per test are flat (3.4–5.3) and whose cost is at most $1.33\,n^2$ (measured 1.12). Two rules have no such machine and cost at most $n^3/685$ and $n^{2.93}$ (both upper bounds, not orders).

We then show why two standard lower-bound techniques fail. The device's answer is one of eight actions, so no information argument can force more than 1.893 reads per query; and no certificate argument can force more than n reads, because certificates do not compose — on one tape the per-query certificates sum to 1 285 703 cells while their union has 1 020. Finally, a bound proved for the rule's output does not bound the walk directly. The walk's cost is the output's cost multiplied by the fraction $f(n)$ of queries at which one answer can change the action: the machine pays only for those queries. Measured, $f = 0.104$ at $n = 128$ and f falls like $n^{-0.99}$, so an output bound of slope s appears as a walk bound of slope $s - 0.99$ — which is why class-restricted bounds restated for the walk carry slopes near 2 rather than 3.

1 Introduction

The object of study is a machine that is almost a two-pointer scan and not quite one. Two headers enclose a window of that tape (three values, or four states with the sign, Section 2); the rule looks

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at the window and names an action; the window grows when the rule moves a header outward and shrinks when it trims a cell off an end. The machine is memoryless — the rule's input is the window's content and the geometry, nothing else — and its state is $O(\log n)$ bits. What makes it interesting is that it is expressive enough to define, on a single tape, rules of very different cost: some are answered in a single read per query, some need the window recomputed, and the difference is not visible in the rule's statement.

The paper's question is the cost of the walk: given a rule, how many cell reads must a machine with $O(\log n)$ bits spend to output δ, as a function of n; the tapes counted are the ones that produce an output. Four things are needed to answer it, and the paper supplies them in this order.

First, a classification. A strategy is a *pointer* strategy if its output can be computed in $O(n)$ reads with $O(\log n)$ bits of state, and a *walk* otherwise; the second word records the absence of a known simulator rather than a proof that none exists. Within the walks, the mean cost the worst-case cost g_ρ and the mean cost e_ρ separate three classes (Section 3); all three are occupied, and a fourth cell that an earlier draft carried — a rule predictable at a linear budget — is empty because $p_\rho = \Theta(e_\rho)$ wherever both are measured with one machine.

Second, the length of the walk. It is not fixed by the model. Measured on the same ensemble at $n = 4096$, the mean step count ranges from 0.385 to 15.58 times $n^2/128$ across the rules, a factor of forty, so there is no rule-independent clock and the cost cannot be factored into a model part and a rule part. The length is carried per rule, and every exponent below is either fitted from the measured costs at four sizes or derived from a step count where the machine's steps are known; measured reads.

Third, a collapse. If the rule's output is a group sum of one-cell terms, three words suffice: a prefix sum maintained incrementally answers every query in one read, and the whole walk in $n + S - 1$ reads (Theorem 1). Six constructions that were once believed to be super-linear are instances of this theorem; each was a machine that recomputed the query, and the rule was linear all along. A second sufficient condition for linear cost is monotonicity (Theorem 1).

Fourth, a threshold. For rules whose output is a kernel correlation, the cost is governed by the kernel's Hankel rank: the reads per query are 1 when the rank fits in the state and m/s when it does not (Theorem 2). Such rules cost $O(n^2)$ or $O(n^3/\log n)$, and nothing in between, and this is the only place in the model where a cubic cost appears.

The last part of the paper is negative, and it is the part that constrains what a proof could look like. Two ceilings bound the two standard techniques: information gives at most 1.893 reads per query (Theorem 3), and certificates give at most n reads in total (Theorem 4), the second because certificates do not compose. And a bound proved for the rule's output must be multiplied by the fraction of queries at which a single answer can change the walk's action, because the machine pays only for those; that fraction is measured, it falls like $n^{-0.99}$, and it is why the class-restricted bounds of the literature, restated for the walk, carry slopes near 2.

Section 2 defines the model and Section 3 gives the classification. Sections 3.1 and 4.1 give the collapse, the monotone theorem and the rank threshold. Section 4 gives the machines, and Section 5 the predictor's budget and the measured constant. Sections 5.1 and 5.2 give the two ceilings and the transfer factor. Section 4.2 collects the costs in one table and states the three conjectures.

2 The machine

Tape. Each cell is an ordered pair $X_i = (b_i^{(1)}, b_i^{(2)}) \in \{0,1\}^2$ — four states per cell, two bits of tape — and the tape is $\tau \in (\{0,1\}^2)^n$. The two columns are drawn independently, each uniformly from the $\binom{n}{h}$ subsets of $[n]$ of size $h = \lceil n/2 \rceil$; tapes are drawn independently of one another. For

even n the four states are equally likely; for odd n, where $h = (n+1)/2$,

$$P(00) = \left(\tfrac{n-1}{2n}\right)^2, \qquad P(01) = P(10) = \frac{n^2-1}{4n^2}, \qquad P(11) = \left(\tfrac{n+1}{2n}\right)^2,$$

and the two columns are negatively correlated.

Sum and label. Write $\sigma_i = b_i^{(1)} + b_i^{(2)} \in \{0,1,2\}$ and $s_i = b_i^{(1)} - b_i^{(2)} \in \{-1,0,+1\}$. The map $X \mapsto (\sigma, s)$ is a bijection: $(0,0) \mapsto (0,0)$, $(0,1) \mapsto (1,-1)$, $(1,0) \mapsto (1,+1)$, $(1,1) \mapsto (2,0)$. The marginal law of σ is $P(0) = P(2) = \frac{1}{4}$ and $P(1) = \frac{1}{2}$ at even n; it is the three-valued measure of the companion paper. A rule stated on σ therefore acts on the sum, and a rule that reads s acts on the full pair. Since $s_i \neq 0$ only where $\sigma_i = 1$, and there $s_i = \pm 1$ equiprobably, the label is a sign carried by the middle level: the four-valued model is the three-valued one with that sign added, and the two agree on every rule that ignores it.

Reads and the charge. A *read* is one access to one cell of the tape; the state's own contents are free. Every machine below is charged by one of two regimes, and every table names which one produced its numbers. Under the *incremental* charge (INC) a machine pays one read for each cell that enters or leaves its window, plus one opening pass over the tape, because a maintained statistic is refreshed from the boundary cell alone; a scan over a window of length m costs m reads. Under the *scan* charge (SCAN) a machine that re-evaluates its window from scratch pays m reads per query. The two regimes differ by a factor of m, so a number carries no meaning without its regime: the tables in Section 4 and Appendix A are INC unless the caption says otherwise, and the scan values kept for comparison are labelled.

State and actions. Fix $n \geq 3$. A region $[L, R]$ and two headers (ℓ, r) with $L \leq \ell \leq r \leq R$, initially $L = \ell = 0$ and $R = r = n-1$. At each step the rule chooses one of eight actions:

$$\ell \mapsto \ell + 1, \qquad \ell \mapsto \ell - 1, \qquad r \mapsto r + 1, \qquad r \mapsto r - 1,$$

$$L \mapsto L + 1 \ \text{(left trim)}, \qquad R \mapsto R - 1 \ \text{(right trim)}, \qquad \text{both (double trim)}, \qquad \text{halt}.$$

The three trims are actions in their own right. They may be taken at any step, not only when a header has reached a boundary, and a rule may trim with both headers far from the ends. Each trim removes one cell from each side it names, so the region only shrinks; after any trim the headers restart at $(\ell, r) = (L, R)$. The tape is read-only.

Definition. A *strategy* is a fixed function

$$f(\varphi(\tau[\ell..r]),\ n,\ \ell,\ r,\ L,\ R) \ \longrightarrow\ \{\ell{+}1,\ \ell{-}1,\ r{+}1,\ r{-}1,\ \text{left trim},\ \text{right trim},\ \text{double trim},\ \text{halt}\},$$

where φ is a fixed function of the window with a finite value set. The geometry is part of the input: a strategy may depend on ℓ, r, L and R as well as on the window. In particular the net inward distance shrink $= (\ell - L) + (R - r)$ is a function of the geometry, so a strategy that counts it is within the model. This is deliberate: the model is a tape machine, not a machine with a step counter, and the distinction is what the two ceilings of Section 5.1 quantify.

A run is defined as follows.

1. Start at $(L, R, \ell, r) = (0, n-1, 0, n-1)$.

2. If $\ell = r$, the run stops and outputs $\delta = \ell/(n-1)$.

3. If the state (L, R, ℓ, r) has occurred before, the run ends with no output: a rule without memory repeats from there, and we stop rather than follow the repetition.

4. Otherwise the rule reads the window, evaluates f, and acts:

   (a) a move is legal only if it keeps both headers inside $[L, R]$, and if it brings them together or past each other the run stops and outputs δ at the position the moving header reaches;

   (b) a trim is legal only if the region keeps at least one cell, and it acts on the region as described above;

   (c) halt ends the run with no output.

Not the companion paper's cut. A trim removes one boundary cell and resets both headers to the ends of the remaining region. It is not the *cut* of the companion walk, which splits the tape at a header's position, advances that header and discards everything outside. The two machines therefore behave differently at a boundary, and the cost of a trim is one cell, not a discarded segment.

What a trim removes. A trim names a side, and the side it names loses one cell. The double trim removes one cell from each side and is a separate action, so a rule that ends an epoch from both sides says so explicitly. Which of the two a rule uses is part of its definition, not of the model, and the choice is visible in its cost: trimming two cells shortens the region twice as fast.

Moving outside the region is not an action. Forcing a header past a boundary is not a legal action, and the model does not repair it: a rule that wants to discard the cell behind a boundary has to say so, and the action that says so is a trim. The rules studied below are written as four-move tables; read as strategies, they choose the trim on the side a move would have left and halt once the region is small. Those are choices of those rules, not clauses of the model.

Outcomes. Three endings. A meeting outputs $\delta = \ell/(n-1)$. A halt outputs nothing. A repeated state outputs nothing.

Cost. Two quantities are used. The *cost* of a rule ρ at n, written $g_\rho(n)$, is the worst-case number of reads of the machine this paper measures for ρ — the e-best one, $M^\star$ — on any tape of length n that produces one; a tape with no output has no δ to reproduce and is not counted. It is a maximum over tapes for that machine, hence an *upper* bound on the min–max over all machines, never a lower one. The tables also report the reads a machine spends per query it answers; that ratio describes the machine that was measured, not the rule. A rule is *fully linear* if $g_\rho = O(n)$ with $O(\log n)$ bits of state, and the ladder below concerns the exponents that appear.

2.1 The length of the walk is a property of the rule

It is tempting to look for a clock — a step count fixed by the model, so that every cost factors as steps times reads per query. There is none. On the same ensemble at $n = 4096$ the mean step count ranges from $0.385\,n^2/128$ for `NBLK>3` to $15.58\,n^2/128$ for `hashmod`, and the growth exponent between $n = 1024$ and 4096 ranges from 1.70 to 2.20; Appendix A.4 records every rule measured.

The variability is not a constant offset: each rule's machine trims at its own cadence, and the cadence fixes the length of the walk.

Two consequences run through the paper. Cost is measured per rule, not read off from the step count, and the decomposition into a model factor and a rule factor is not available. What survives is the comparison of rules at fixed n, which is what the tables report.

3 Three costs, and the classification

A strategy of this model is a window function φ together with a decision function f that sees φ, both header positions and both boundaries, and returns one of the eight actions of Section 2. What separates strategies for our purposes is not what they read but what their output costs.

Definition 1 (Pointer and walk)**.** *A strategy is a* pointer *strategy if its output can be computed in $O(n)$ cell reads with $O(\log n)$ bits of state — that is, if we possess such a simulator. We use* walk *for the rest, and the word records the absence of a simulator we know rather than a proof that none exists: Section 5.1 says what a proof would need.*

The classification is operational, and every claim below is a construction, a simulator with a read count, unless it says otherwise. The second half of the definition is negative, so a rule is placed on the expensive side only with the evidence we have: a simulator whose cost is $\omega(n)$ and no $O(n)$ one. We write "no $O(n)$ simulator is known" where that is all we can say.

Two costs and a prediction cost. For a rule ρ let

$$G_\rho(n) = \exp \mathbb{E}_\tau\big[\ln C_{M^\star}(\tau)\big], \qquad e_\rho(n) = \mathbb{E}_\tau\big[C_{M^\star}(\tau)\big], \qquad g_\rho(n) = \max_\tau C_{M^\star}(\tau),$$

all on one machine, $M^\star$, the e-best of those built, and all in cell reads with $O(\log n)$ bits of state. The rule's own worst case is the min–max over machines, $g_\rho^{\mathrm{mm}}(n) = \min_M \max_\tau C_M(\tau)$; since $M^\star$ is one machine, $g_\rho^{\mathrm{mm}} \le g_\rho$, so every reported worst case is an *upper* bound on the rule's. "Fully linear", $g_\rho = O(n)$, therefore implies $g_\rho^{\mathrm{mm}} = O(n)$, but not conversely: a rule whose optimal machine is not the one we built could be placed too cheaply. Let $p_\rho(n)$ be the per-tape cost at which a predictor with $O(\log n)$ bits drives the out-of-fold residual to zero. The worst case separates the fully-linear class from the rest, and the mean separates the semi-linear class from the superlinear one. The prediction cost p_ρ is not a third separator: it equals e_ρ to within a constant wherever both are measured with one machine. Inside the superlinear class the measured predictability κ_ρ of Section 5 splits weak ($0 < \kappa < 1$) from strong ($\kappa = 0$).

class	in words	criterion	examples
fully linear	cheap wherever it outputs	$g_\rho = O(n)$	monotone family, `nhalf`, `pattern`, `runs`, `peak`
semi-linear	cheap on average, expensive on a vanishing set of tapes	$e_\rho = O(n)$, $g_\rho = \omega(n)$	`ballast`, constructed (Section 3)
superlinear	expensive	$e_\rho = \omega(n)$; weak if $0 < \kappa_\rho < 1$, strong if $\kappa_\rho = 0$	seventeen natural rules; three mechanisms, below

Table 1: The three classes, read as an ordered decision procedure: test $g_\rho = O(n)$; if it fails, test $e_\rho = O(n)$; if that fails the rule is superlinear, and it is split by κ_ρ into weak ($0 < \kappa_\rho < 1$) and strong ($\kappa_\rho = 0$). The cases are ordered and exhaustive, so every rule falls in exactly one, and all three are occupied.

The classification is a definition. The three classes of Table 1 are separated by an ordered decision procedure. Given a rule, test in turn: is $g_\rho = O(n)$? then it is *fully linear*; else, is $e_\rho = O(n)$? then it is *semi-linear*; else *superlinear*, split by κ_ρ into weak ($0 < \kappa_\rho < 1$) and strong ($\kappa_\rho = 0$). The cases are ordered and exhaustive, so every rule falls in exactly one and none escapes: exhaustiveness is a property of the definition, not a claim about rules. Here p_ρ is the per-tape cost at which a predictor with $O(\log n)$ bits drives the out-of-fold residual to zero. It is a cost, not a global budget: a single budget that reaches $R^2 \geq 0.99$ equals the 99th percentile of the per-tape cost, so that quantity is a quantile of g_ρ and not a third measure.

What is a claim. Occupancy is — for three cells, and all three carry members. The fully-linear cell holds four natural rules and the half-power construction; the superlinear cell holds the remaining seventeen; the semi-linear cell held nothing until it was constructed, and Section 3 gives the construction. What is worth stating is why a fourth cell that an earlier draft carried was removed: the measure that was to separate it, p_ρ, turns out to equal e_ρ to within a constant on every rule where the two have been measured with the same machine, so no rule with $e = \omega(n)$ and $p = O(n)$ exists in the catalogue. That is a measurement with a mechanism behind it: a predictor that drives the residual to zero on all but a vanishing fraction of tapes is already a machine on those tapes.

What the measurements place. The fully-linear cell holds four rules, each with an exact machine whose state is $O(\log n)$ bits and whose reads are a constant per cell of the tape rather than a window scan per query: the monotone family ($e_\rho = g_\rho = 2n - 1$ by construction), `pattern` (3.7–4.2 n, exact on every one of 200 meeting tapes at $n \leq 2048$ and 100 at $n = 4096$), `runs` (3.8–8.5 n for $n \geq 1024$, exact on 100 per cent of the meeting tapes) and `peak` (2.0 n at 1 496–1 497 of 1 500 meeting tapes; exactness on every tape needs an endgame buffer of width 0.75 n, which costs about 7 n on a 20-tape sample at $n = 2048$). The same question — is there a machine exact on the meeting tapes with $O(n)$ reads? — was then put to the other fifteen rules, and no machine in the family we tested has both properties. The failures fall into three mechanisms, each measured rather than inferred: the epoch law fails in mid-tape on a 0.06–17 per cent set of epochs (`dense2`, `trip`, `oddrun`, `canonical`, `MAXP`); the statistic slides, so the level query pays $\Theta(m)$ per epoch and the cost is $n^2/4$ to within six per cent (`BAL>m/4`, `NBLK=1`, `NBLK>2`, `NBLK>3`, all exact on 700 of 700 tapes); or the terminal is long, reaching $M_0 = 820$ (`AMAX`) and 616 (`AMIN`) at $n = 2048$, which forces an endgame buffer of width $\Omega(n)$, and a buffer of width $\Theta(n)$ suffices and hence $\Theta(n^2)$. The

decisive split is the terminal distribution: mispredictions with $M_0 \le 256$ are at least 91 per cent of the total for every rule that comes out exact and at most 79 per cent for every rule that does not. Two claims about the mean must be withdrawn. First, there is no linear bulk in e: at $n = 4096$ the tapes costing more than $100\,n$ contribute 73 to 98 per cent of the mean (the $10\times$-median cut gives 48 to 97), so the median ($5.3\,n$ for `BAL>m/2` and `KCHG`, flat in n; the eleven-rule range is 4.1–145) and the mean are different objects and the mean is the tail. Second, the order of e is not determined by our data. Writing $e \approx f \cdot C$ with C grows as n^2 (fitted) with a tail weight $f \sim n^{-a}$, giving $e \sim n^{2-a}$, so an intermediate exponent is a stable possibility rather than a crossover; but the measured a is 0.08 to 0.98 depending on how the tail is defined, and the local exponent of e over $n = 512$ to 4096 ranges from 1.05 to 2.76 across consecutive intervals without converging. A heavy-tailed mean at these sizes does not fix an exponent: what the data support is $e = \omega(n)$, and no more.

A third order? The three measures are ordered $p_\rho \le e_\rho \le g_\rho$, and the question this raises is whether both inequalities can be strict at once — a rule with $p_\rho = o(e_\rho)$ and $e_\rho = o(g_\rho)$, whose three costs are in three different orders. The two separations pull in different directions and that is what makes the question hard. For a predictor of cost C that errs on a fraction ε of the meeting tapes, patching it with a fallback of cost M gives an exact machine with mean at most $C + \varepsilon M$; so $p = o(e)$ requires $\varepsilon M \gg p$, while $e = o(g)$ requires the errant tapes to be the worst tapes. A rule whose costs separate three ways must therefore have a defect that is rare, expensive, *and* invisible to a cheap predictor at the same time.

We have no such rule. A construction aimed at it — a defect defined by a hash of the window, a predictor that samples $\sqrt{n}$ cells and misses the defect, an exact machine that reads the full window once per query — was obstructed three ways in `probe_three_order2`. The defect fraction did not decay with n (the share of tapes costing more than $300\,n$ rose from 0.21 to 0.64 over $n = 512 \to 4096$), so the εM lever had nothing rare to grip; e's and g's exponents moved together across the band family ($g/e = 3.4$–6.2 across $K = 2$–16); and the predictor, once it works on all but a vanishing fraction of tapes, is already a machine on those tapes, which drags e down to its order. The measured separations are one, but the structure says there should be many: every superlinear rule's mean is tail-dominated (48 to 97 per cent of e at $n = 4096$ under the $10\times$-median cut, 73 to 98 under $100\,n$) with a near-tail index of 0.55 to 1.9 at the 25 per cent depth, and a heavy tail under a maximum at $\Theta(n^2)$ forces $e = o(g)$; what stops us from exhibiting the separations one by one is that a heavy-tailed mean does not converge at these sizes (Section 3). `longrun` is not the exception but the extreme: $e \ll g$ with $g/e = 86\,232$ at $n = 4096$, growing like $n^2/\log n$, from the structured witness $2^{n/2}0^{n/2}$, not from a three-way split. We therefore state the existence of a three-order rule as an open problem, with the obstruction above as the reason to expect that it has none.

Measuring g. The quantity g_ρ is a maximum over an ensemble whose support has $\exp(\Theta(n))$ tapes, and a uniform sample of 300 to 6000 tapes essentially never draws the structured tapes that maximise the walk. A witness panel of nine structured tapes, each built as a pair of balanced columns so that it lies in the support, lifts g at $n = 512$ by factors of 1.3 to 274 for six of the eighteen rules in the panel; the lifted values have clean exponents ($g = n^{2.00\text{–}2.02}$ for fourteen of the eighteen rules in the panel, against $n^{-0.36\text{–}2.17}$ for the sample maxima), so the panel is the better estimator where it applies. For `canonical` the reverse holds: its worst tape found so far is generic — 311 runs of length 1 to 8 with counts 132/248/132 at $n = 512$ — so the panel does not find it and the sample does. Both estimators are therefore needed, and a statement of the form $g_\rho = \Theta(e_\rho)$ is a statement about the estimator as much as about the rule.

Three placements of the earlier draft moved. `canonical` is not fully linear: under the model's own trim it costs $147.7\,n$ reads at $n = 1024$ under the `pcr_lib.Runner` device (the `c1a` variant gives $116.5\,n$; the two differ by 27%, so the machine must be named with the number), and its step count measures as n^2; the $5.3\,n$ of the earlier draft is the meeting-conditional cost of a region-collapse variant, not of this model. `hashmod` is not predictable at a linear budget: its mean is 0.08–$0.096\,n^2$ cell reads at $n = 2048$ (the $0.62\,n \ln n$ of the record counts *rows*, not cells, and is 4041% away from the cell count; the audit in `probe_master/audit2.py` catches this), and its worst case is $\Theta(n^2)$, and under a linear budget its residual does not vanish. `longrun` is not semi-linear: its mean is $7.02 + 0.15 \log_2 n$ reads per cell over $n = 512$–16384 (asymptotically $4.8 + 0.31 \log_2 n$), so it grows like $n \log n$, and its worst case is the structured tape $2^{n/2}0^{n/2}$ at $n^3/21$ reads — measured at four sizes (6 544 735, 50 408 550, 395 496 223, 3 132 928 486 for $n = 512, 1024, 2048, 4096$, i.e. $n^3/20.5$, $1/21.3$, $1/21.7$, $1/21.9$); the "$25\,n^2$" of an earlier draft is the $n = 512$ slice of that sequence, whose n^2 coefficient doubles at every doubling of n. So `longrun`'s costs separate by $n^2/\log n$: $g/e = 86\,232$ at $n = 4096$.

Discreteness is not established. A rule that declares its own read length $L = m^\sigma$ has the upper bound $\sum_q L = n^{2+\sigma}(1 + o(1))$, but that is an identity about the declared reads, not a lower bound on the cost: the family we tested attains between 0.76 and 1.00 of the ceiling, and a 256-entry cache — legal at $O(\log n)$ bits — reads 0.0019 of it at $\sigma = 0.5$. The exponents of this paper are therefore a list of mechanisms, not an enumeration of what the model admits, and we claim neither that the exponent set is discrete nor that it is continuous.

3.1 Linear rules, and the collapse

What a group sum costs. The cheapest rules are those whose output is a group sum of one-cell terms. For these the window's value is a prefix-sum difference, and a machine that maintains the prefix sum incrementally pays one read per query.

Proposition 1 (Collapse, verified)**.** *If φ is a group sum of one-cell terms, there is a machine with three words that answers every query in exactly $n + S - 1$ reads, where S is the number of steps — for the rules whose window statistic is a sum over the window carried incrementally, which is the class the proposition is stated for; a statistic whose value is not a window sum is outside it.*

The proposition was checked on 320 880 queries with no error — every tape of the cube up to $n = 10$ and random tapes to $n = 96$ — and it is the reason six constructions had to be withdrawn. The general statement is not proved here:

Conjecture 1 (The collapse at general n)**.** *The bound $n + S - 1$ holds for every n and every group-sum rule. No counterexample exists in the tested range and the mechanism is understood, but the passage from the tested cases to all n is open.*

The six constructions are the phase family, the (n, n^2) family, the perfect-square rule, the run-length rule, the level band, and Rule B were each announced as super-linear and each turned out to be a machine that recomputed the query. None was a property of its rule.

3.2 The monotone theorem

Theorem 1 (Monotone rules are predictable)**.** *Fix a strategy (φ, f) with f monotone: every move is $\ell \mapsto \ell + 1$ or $r \mapsto r - 1$, no trim action and no halt action is ever chosen, so exactly one cell leaves the window per step. Suppose further that the predictor may read the initial window, and*

that the state s_0 determining φ on it is computable from that window in $O(n)$ reads with $O(\log n)$ bits of workspace — a hypothesis that a finite value set of φ does not by itself supply. Suppose φ is incrementally computable*: there is a machine M with $O(\log n)$ bits of workspace which, given the window $[\ell, r]$, a state s that determines $\varphi(\tau[\ell..r])$, and the cell that is about to leave, produces in $O(1)$ steps a state s' that determines φ on the new window. Then there is a predictor with $O(\log n)$ bits of workspace that computes δ in at most $2n$ reads.*

Proof. The predictor keeps (ℓ, r) and s; initially it reads the whole window, n reads, and computes s_0 by the hypothesis just stated. At each step it evaluates φ from s, applies f, and calls M with the cell that leaves. By induction on the step the predictor's (s, ℓ, r) equals the rule's at every step: the base case is the initial window, and the step case is the definition of M. Hence the first time the headers would coincide is the same for both, and the predicted δ equals the rule's. One cell is read per step, so the total is n for the initial window plus at most $n - 1$ for the steps. □

The hypothesis is met by the two features of the canonical rule separately. The three window sums $S_1 = \sum a_i$, $S_2 = \sum a_i^2$, $S_3 = \sum a_i a_{i+1}$ are updated by subtracting one term, and φ can be evaluated from them; the mean of the first half is updated by moving its two ends by at most one cell each. Both features are therefore maintainable along a move, and a monotone rule that reads them is cheap by the theorem.

Scope. The theorem covers monotone rules only. The canonical rule is not one: two of its four moves enlarge the window. The bound of $2n$ reads therefore says nothing about it, and the machines of Section 4 are not a counterexample to the theorem but a statement about a rule outside its hypothesis.

A closed form for comparison. One walk of this model has a closed form, and it is not monotone: with $b_1 \equiv 0$ the canonical table becomes a largest-concavity scan. Starting from the region $[L, R]$, let

$$m^\star = \max\{m \in [2, R - L + 1] \ : \ A_{\lfloor m/2 \rfloor} \geq A_m\},$$

with $m^\star$ undefined if the set is empty, where A_m is the mean of the first m cells of the region. If $R - L + 1 < 3$ the run halts; if $m^\star$ is undefined the headers meet and the output is $L/(n-1)$; if $m^\star = R - L + 1$ the region is trimmed on both sides; otherwise it is trimmed on the left. A run therefore performs at most $\binom{n+1}{2}$ tests of b_2, hence $O(n^2)$ steps.

The theorem is a sufficient condition for membership in the fully-linear class. The conjectures of Section 4.2 concern the rules outside its hypothesis.

The canonical rule. The rule that motivates the model is a four-move table on the three values: it reads a window of length m, computes two bits, and moves a header or trims accordingly. Under the model's own trim its cost is $147.7\,n$ reads at $n = 1024$ under the `pcr_lib.Runner` device, and its step count grows like n^2: the walk is long, and the machine that follows it pays for that. With $O(n)$ bits of state — outside the model, whose budget is $O(\log n)$ — the same rule costs exactly n reads, one pass over the tape, so the $147.7\,n$ is the price of the state budget, not of the rule. The $5.3\,n$ quoted in an earlier draft is the meeting-conditional cost of a region-collapse variant whose trims take two cells at a full window; it is not this model.

The classification of Section 3 uses a mean cost and a worst-case cost. There is a third cost between them, and it is the one that decides whether a rule deserves to be called *perfect*: the geometric mean of the same read counts.

The three costs

Fix a rule ρ and a size n, and let $\mathcal{M}_\rho(n)$ be the machines that are exact on every meeting tape. Write $C_M(\tau)$ for the number of cells M reads on the meeting tape τ, and let $M^\star$ be a machine attaining $e_\rho(n) = \min_{M \in \mathcal{M}_\rho(n)} \mathbb{E}_\tau C_M(\tau)$. Define, on that one machine,

$$G_\rho(n) \;=\; \exp\big(\mathbb{E}_\tau \ln C_{M^\star}(\tau)\big), \qquad e_\rho(n) \;=\; \mathbb{E}_\tau C_{M^\star}(\tau), \qquad g_\rho(n) \;=\; \max_\tau C_{M^\star}(\tau). \tag{1}$$

All expectations and maxima are over the meeting tapes of T_n. Measuring the three on one machine is what makes them comparable. e_ρ and g_ρ are the mean and the maximum of the same machine $M^\star$, the best we have; neither is an infimum or a min–max over all machines. In particular $g_\rho = \max_\tau C_{M^\star}(\tau) \geq \min_M \max_\tau C_M(\tau)$, so every worst-case number reported here is an upper bound on the true min–max.

Proposition 2 (The chain)**.** *For every rule and every n,*

$$G_\rho(n) \;\leq\; e_\rho(n) \;\leq\; g_\rho(n). \tag{2}$$

Moreover $G_\rho = e_\rho$ if and only if the cost $C_{M^\star}$ is constant on the meeting tapes.

Proof. The first inequality is AM–GM applied to the positive numbers $\{C_{M^\star}(\tau)\}_\tau$: $\exp(\mathbb{E} \ln C) \leq \mathbb{E}C$, with equality exactly when they are all equal. The second is $\mathbb{E}C \leq \max C$. □

Proposition 3 (Spread bound, by definition)**.** *Let $\chi = \chi_\rho(n)$ be the ratio of the largest to the smallest cost of $M^\star$ on the meeting tapes — a ratio of costs, written χ to keep it apart from the predictive κ of Definition 10. Then, because G_ρ and e_ρ lie between the two extreme tapes,*

$$\frac{g_\rho}{G_\rho} \;\leq\; \chi, \qquad \frac{g_\rho}{e_\rho} \;\leq\; \chi, \qquad \frac{e_\rho}{G_\rho} \;\leq\; \chi. \tag{3}$$

In particular, if $\chi = O(1)$ then $G_\rho = \Theta(e_\rho) = \Theta(g_\rho)$: the rule is perfect*.*

Proof. Write $c_- = \min_\tau C_{M^\star}$ and $c_+ = \max_\tau C_{M^\star}$, so $\kappa = c_+/c_-$. Every cost lies in $[c_-, c_+]$, hence so does every mean of them: $c_- \leq G_\rho \leq e_\rho \leq c_+$. Dividing by $G_\rho \geq c_-$ gives $g_\rho/G_\rho \leq c_+/c_- = \kappa$; dividing $c_+ \leq \kappa c_-$ by G_ρ, e_ρ gives the other two. □

Corollary 1. *A rule is perfect if its e-best machine has a bounded cost ratio. This covers two families.*

1. Tape-free costs. *If $C_{M^\star}$ depends on n alone then $\kappa = 1$ and $G_\rho = e_\rho = g_\rho$ exactly. This is the case for the monotone family ($2n - 1$), the half-power rule ($\lfloor \sqrt{n-1} \rfloor + 2$), the grid family,* `DC` *and* `PRF`*, whose measured g/e is* 1.000*.*

2. Bounded measured spread. *Measured on the CES device at four sizes,* `pattern` *has $G_\rho/e_\rho =$ 0.997–0.998 and $g_\rho/e_\rho \leq 1.11$, and* `runs` 0.997 *and* ≤ 1.16*: the measured $\chi \leq 1.16$ then puts all three of their costs within sixteen per cent of one another. This is a measurement about the tapes, not a theorem about rules: the proposition itself is an identity, and its content is that κ is small on this catalogue.* `peak` *is measured on the same footing: G_ρ/e_ρ rises* $0.984 \to 1.000$ *and g_ρ/e_ρ falls* $2.03 \to 1.05$ *over* $n = 512 \to 4096$*, so it is perfect as well.*

What the third cost shows

With G measured, the catalogue splits cleanly. For the band family at $n = 4096$,

$$\frac{G_\rho}{e_\rho} = 0.08\text{–}0.26, \qquad G_\rho = n^{1.09\text{–}1.90}, \quad e_\rho = n^{1.49\text{–}2.15}, \quad g_\rho = n^{2.00\text{–}2.02}. \tag{4}$$

The three exponents are distinct, so none of these rules is perfect. For the fully linear rules the three collapse: $G_\rho = \Theta(e_\rho) = \Theta(g_\rho) = \Theta(n)$ by Corollary 1 — the measured means are $2n - 1$, $\lfloor\sqrt{n-1}\rfloor+2$, $2.57\,n$, $2.77\,n$ and $2.0\,n$ for `monotone`, `nhalf`, `pattern`, `runs` and `peak`. In the fourteen content-dependent rules measured, every perfect one is at most linear and every superlinear one has $G_\rho/e_\rho \le 0.26$.

rule	G_ρ/n	G exp	e_ρ/n	e exp	g_ρ/n^2	g exp
`AMAX`	31.68	1.55	383.8	2.13	1.12	2.02
`AMIN`	29.66	1.40	305.9	1.85	1.12	2.02
`BAL>m/2`	10.64	1.17	42.8	1.62	1.25	2.00
`BAL>m/4`	12.92	1.18	52.7	1.56	1.25	2.00
`KCHG`	9.62	1.09	37.0	1.49	0.75	2.00
`LAS`	41.69	1.48	320.7	1.97	0.75	2.00
`MAXP`	53.33	1.43	347.2	1.89	0.75	2.00
`NBLK=1`	22.61	1.44	134.1	1.81	0.87	2.01
`NBLK>2`	9.91	1.32	43.6	1.79	—	—
`NBLK>3`	8.95	1.32	39.0	1.83	—	—
`oddrun`	102.19	1.90	502.8	2.15	0.75	2.00

Table 2: The three costs for the band family, $n = 4096$ (values) and log–log exponents over $n = 512 \ldots 4096$. G and e are measured on the same machine, so Proposition 2 applies row by row; g_ρ is the maximum over the nine-tape witness panel, in units of n^2. `NBLK>2` and `NBLK>3` have no witness in that panel. Every cost is read from the `hmach3.H3` band device at $B = 64$, charged one read per cell taken or given up plus the opening pass. The exponents are log–log slopes over $n = 512 \ldots 4096$; a 400-draw bootstrap that resamples the tapes at every size gives 95 per cent intervals of ± 0.2 on the e column and ± 0.1 on the G column, and the e and G intervals are disjoint in ten of the eleven rows and touch at 2.04 in `oddrun` (G $[1.75, 2.04]$, e $[2.04, 2.26]$), so the separation between the columns is not a sampling artefact except at that one boundary. No row has matching exponents: all eleven are non-perfect.

Two statements that are not theorems

Sublinear does not imply perfect. One can build a rule whose cost is 0 reads on a $1 - n^{-2}$ share of the meeting tapes (its output is then determined by the geometry) and n reads on the rest, with a hash-defined defect deciding which. Then $e_\rho = O(1)$ is sublinear, $g_\rho = \Theta(n)$ and $G_\rho = O(1)$, so $G_\rho \ll g_\rho$: sublinear, not perfect. The conjecture below therefore has to be restricted to content-dependent rules and stated in the direction that survives.

Conjecture 2 (No superlinear perfect rule)**.** *Every content-dependent perfect rule is at most linear: there is no content-dependent rule with $e_\rho = \omega(n)$ and $G_\rho = \Theta(e_\rho) = \Theta(g_\rho)$. The spread is a separate question: e_ρ/G_ρ does take intermediate values — the eleven band rules sit between 4 and 12 — so the conjecture is the statement that no content-dependent rule is perfect, and not a statement about the range of the spread.*

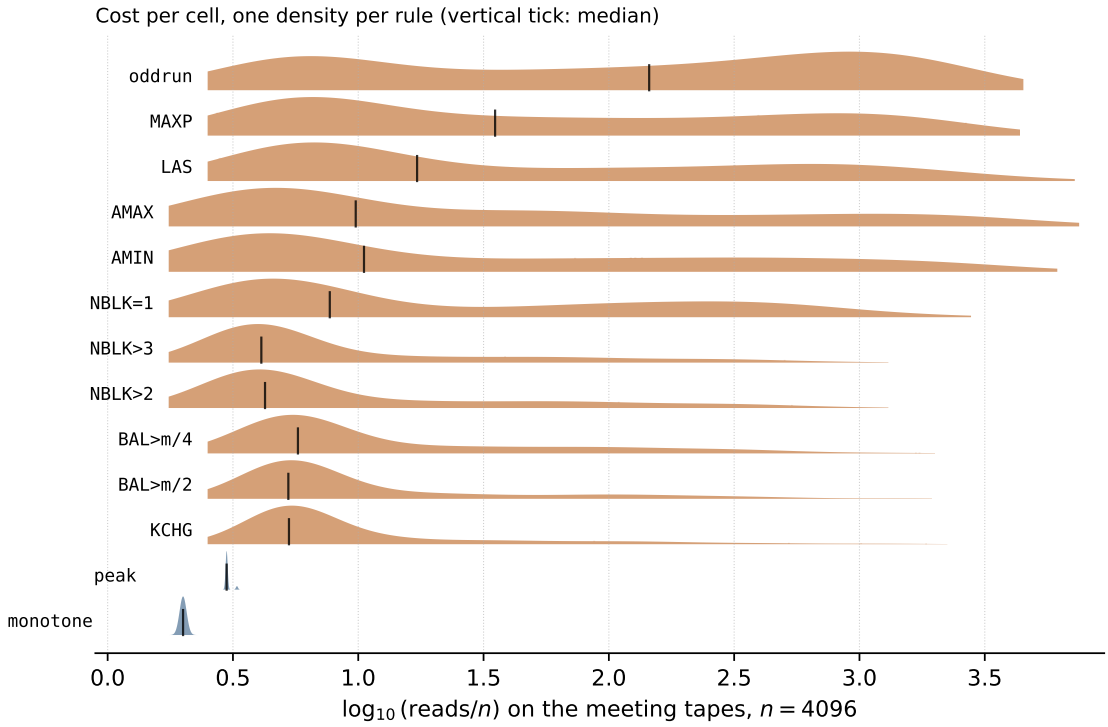


Figure 1: The cost per cell of each rule's e-best machine on the meeting tapes at $n = 4096$, one density per rule on a log scale, the vertical tick marking the median. The fully linear rule at the top is a spike at $2n$: every tape costs the same. The eleven band rules below spread over two to four decades, with their mass in the right tail — the median is 5 to $145\,n$ while the ninety-ninth percentile reaches $2\,200\,n$ — which is why their means sit far above their medians and why G, e and g separate on them. `peak` is omitted: the per-tape reads stored for it come from a twelve-tape halted sample that disagrees with the 1496/1500 figure in Section 4.

The evidence is the eleven superlinear rows of Table 2, whose e_ρ/G_ρ is 3.85 to 12, together with the three perfect rules of Section 3, all of which have $e_\rho = \Theta(n)$. A counterexample would be a rule whose e-best machine is exact with a bounded cost ratio and yet costs $\omega(n)$; the constructed tape-free families show that such a machine is not forbidden by the model, only by content-dependence.

The semi-linear cell: `ballast`. On the natural rules the cell is empty, so it is occupied by a constructed rule, and a construction is what the cell needs: nothing in the catalogue has a linear mean with a quadratic worst case, and without a member the class would be a definition with no witness. `ballast` is built from two branches, and both design choices are the opposite of the obvious one.

Its normal branch is a monotone collapse: exactly $n - 2$ steps, no trim, one read a step. That branch is *not* a constant action — the side it takes is a bit of the window — so the output moves by a third of the tape from one tape to the next (sd $0.35\,n$ cells). The cost is linear and the answer is content-dependent, which is what keeps a zero-read machine from existing for it.

Its defect is a one-shot property of the whole tape, read once at the opening. It fires with

probability $1/n$, and the branch it opens costs $0.75\,n^2$ reads. The mean is therefore

$$5.44\,n \;+\; n^{-1}\cdot 0.75\,n^2 \;=\; 6.2\,n,$$

linear because the rare branch is rare enough to be absorbed, while the worst tape pays $0.75\,n^2$. Measured over $n = 512 \ldots 4096$ on 200 meeting tapes per cell, one machine: $e \sim n^{0.997}$, $g \sim n^{1.995}$, $G \sim n^{1.000}$, with e/n flat at 6.11–6.24 and g/n^2 at 0.751–0.759. The mean is at n, the worst case at n^2, and the spread between them is what makes the class non-empty. The rule table is `sl2_engine.walk2`, rule `R_BALLAST`, in the deposit.

4 The machines and the ladder

Five level-count rules. The rules below ask a question of the form "how many cells of the window lie at a given level", which is not a group sum and so is outside the collapse. Each has a machine that keeps a band around each header plus a frontier of counters, and each was measured with the reads per test flat over a doubling of n.

rule	question	machine	reads per test
`NBLK>2`, `NBLK>3`	count ≥ 3 or ≥ 4	band + frontier, $B = 64$	4.36–4.50
`NBLK=1`	count $= 1$	band + frontier, $B = 64$	3.44–3.51
`AMAX`	endpoint is the strict max	band + frontier + certifier	4.79–5.26
`AMIN`	endpoint is the strict min	band + frontier + certifier	4.41–4.87

Table 3: Measured at $n = 4096, 8192, 16384$; the range is over the three sizes. The cost law $e = a \cdot \text{tests} + b \cdot \text{epochs}$ fits with a residual below one per cent and a flat in n.

The four rules cost at most $1.33\,n^2$ reads (measured 1.12) on the band + frontier device, charged one read per cell taken or given up plus the opening pass, with the constant measured on scale-free tapes (0.655 for `NBLK=1` on ramp tapes, 0.87 for `NBLK=1` on plateau tapes, 1.12 for the two extrema) and constant to 0.6 per cent over $n = 1024$ to 8192. The measurements were verified at every query: 416 453 444 queries re-derived and compared with zero mismatches, and 7 793 964 scan events priced by the certifier with zero mismatches.

Two rules without such a machine. The pointer-chase rule (`PC`) and the kernel correlation rule (`CX`) have no band machine. For `PC` the best known machine scans from the right end at each test and costs $n^3/685$ on the tape 1^m with nine defects, a fixed-defect-count family whose local exponent is 2.90; with a fixed defect *density* the same machine costs between $n^{2.5}$ and $n^{2.85}$. For `CX` the best known machine is an s-slot accumulator, measured at $n^{2.93}$ with a fixed slot count; that is an upper bound, not the order, and the rank of its kernel is exactly m, so Section 4.1 applies and gives $n^3/\log n$ once $s = \Theta(\log n)$.

4.1 The rank threshold

Let h be the kernel of a correlation rule, so that the query at a window of width m is $\varphi(\ell) = \mathbb{1}[\sum_i \sigma_i h(i-\ell) \bmod q < q/2]$, and let $H_m[d,k] = h(k+d)$ be its Hankel matrix over $\mathbb{F}_q$. By Kronecker's theorem the minimal recurrence order of the kernel is the rank of H_m.

Theorem 2 (Cost of a kernel rule)**.** *For a kernel-correlation rule,* 1 *read per query suffices when* $\operatorname{rank}(H_m) \leq s$*, and* $\lceil m/s \rceil$ *reads per query suffice when it exceeds* s*, where* s *is the number of*

linear functionals the state holds. The budget allows $s = O(\log n)$; the machines of this paper take $s = \Theta(\log n)$, and at that instance a full-rank kernel costs $O(n^3/\log n)$ reads over the $\Theta(n^2)$ queries of the walk. Both bounds are upper bounds on what a machine needs; the matching lower bound is Conjecture 3, and it is not proved here.

Conjecture 3 (The per-query floor for kernel rules)**.** *The $\lceil m/s \rceil$ above is necessary as well: no machine whose state holds s linear functionals answers a full-rank kernel query in fewer than $\lceil m/s \rceil$ reads, so no kernel rule with a full-rank kernel has cost $o(n^3/\log n)$. This is the step from s functionals to $\lceil m/s \rceil$ cell reads. It is the reason the $n^3/\log n$ entry in the ladder is an upper bound with a conjecture under it rather than a tight order.*

For the convolution rule the rank is exactly m: $H_m = 3\,\mathrm{diag}(3^{d^2})\,[9^{dk}]\,\mathrm{diag}(3^{k^2})$ is a Vandermonde matrix whose nodes are distinct because $\mathrm{ord}_q(9) > m$ for every m in range. This is why that rule sits at the top of the ladder: its kernel has full rank, so no fixed-dimension linear state can summarise it, and every machine must either spend m reads per query or hold m functionals.

4.2 The ladder

rung	rule	machine	mechanism
$n^{1/2}$	`nhalf`	one defect after the first trim	$g = \lfloor\sqrt{n-1}\rfloor + 2$, both boun
n	monotone family	counter	Theorem 1
$n^{3/2}$	constructed	pursuit + monotone walk	measured
n^2	`NBLK>2`, `NBLK>3`, `NBLK=1`, `AMAX`, `AMIN`	band + frontier, $B = 64$	$\le 1.33\,n^2$ (measured 1.12), co
$n^{5/2}$	`PC`	Brent cycle detection	orbit length $\Theta(\sqrt{m})$; $n^3/685$
$n^3/\log n$	`CX`	$A = s/31$ anchors	$\Theta(n^3/\log n)$ for linear machi
n^3	`PRF`; `longrun`	one window read per test; a witness tape	$\Theta(n^3)$ at $O(\log n)$ bits; $n^3/21$

Table 4: The ladder of mechanisms, with the device that attains each rung named in the third column. The rungs are 1/2, 1, 3/2, 2, 5/2, 3 and 3 with a logarithm; the logarithm is a division, not a multiplication, because the anchor machine divides a full-window scan by $A = s/31$ and the batch machine divides it by $r = \sqrt{2s/q}$, and the two do not compose (Section 4). The `PRF` entry in the n^3 row is conditional on the PRF heuristic; the `longrun` entry is an *upper* bound, measured on its witness tape at four sizes (the coefficient of n^3 moves only from 1/20.5 to 1/21.9), not a lower bound. The $n^3/\log n$ row is proved for linear machines.

Read as a ladder, the table has rungs at 1/2, 1, 3/2, 2, 5/2, 3/log and 3: the group-sum rules (Theorem 1) at n, the constructed half-power rule, the five level-count rules at n^2, the pursuit rule at $n^{5/2}$, the kernel rule at $n^3/\log n$, and — conditionally on the PRF heuristic, and only so — the one rule at n^3, joined there through its witness by `longrun`.

Is the top rung reachable? `CX` has a machine at $n^3/\log n$ and a matching lower bound for linear state; `PRF` has a machine and a lower bound at n^3, the latter conditional on the PRF heuristic. So the ladder reaches n^3 and the logarithm on the row below it is a property of `CX`'s algebra, not of the model: one 31-bit residue can be updated in one read per right move, which is what buys the factor $\log n$ and nothing more.

Conjecture 4 (The exponent set)**.** *For every rule ρ the worst-case exponent of g_ρ lies in $\{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3\}$, with a logarithmic correction admitted on the top rung only — the $n^3/\log n$ of the kernel rule; a factor $n^{1/\log n}$ would be a constant and is not a rung.*

Conjecture 4, and its limits. The earlier form of the statement was the set $\{0, 1, 2, 3\}$, and it is now wrong in two places: the half-power rule `nhalf` is a proved member of the ladder, and `CX` carries a logarithm. What survives is the absence of any fractional exponent other than $1/2$ and $5/2$, and that is a statement about the rules examined, not about the model: a rule that declares its own read length $L = m^\sigma$ has an upper bound $n^{2+\sigma}$ for every σ, and while the family we tested attains only 0.76–1.00 of that ceiling, no lower bound rules the intermediate exponents out.

The classification, and where it stands. The classification itself is definitional (Section 3), so no rule can escape it; what is a claim about rules is which cells are occupied. All three are: the semi-linear cell is occupied by the construction of Section 3, the fully-linear one by four natural rules and the half-power construction, and the superlinear one by the remaining seventeen. Conjecture 4 (above) asserts the exponent set $\{0, 1, 2, 3\}$; it is consistent with every measurement here, and the one apparent counterexample has been withdrawn. Conjecture 5 (Section 5) asserts a single threshold $\kappa^\star < 1$ between the weak-superlinear band and full linearity; under the incremental charge, at $c = 16$ and $n = 512$, the thirteen rules measured span 0.472 to 0.925 (`NBLK>2` 0.826, `NBLK>3` 0.819, `canonical` 0.925) while `hashmod` is at approximately 0, which bounds $\kappa^\star \geq 0.93$ without locating it. With no rule sitting between the measured band and $\kappa = 1$, the threshold no longer has an upper witness, and the conjecture is left with its lower end only.

5 Limits: what cannot be proved

A ceiling is a statement about what a predictor can recover, so it needs two things the earlier sections do not supply: a budget, and a measured constant. This section fixes both, and states the one conjecture of the paper that is about prediction rather than about cost.

The predictor. A predictor is charged a budget B cell reads and $c \log_2 n$ bits of state, pays m reads per step of the walk it follows, answers exactly the tapes it finishes inside the budget, and predicts the rest from the checkpoint state. R^2 is out-of-fold, so $R^2 = F + (1 - F)\rho$ with F the finished fraction; κ_ρ is its limsup in n. The budget grid, the register count and the fold structure are recorded in Appendix A.2, with the per-rule table at $c = 16$, $n = 512 \to 4096$: five rules, all falling in n, the largest 0.999 at $c = 128$. The paragraph below records why the charge matters and what the numbers do and do not bound.

The charge is the whole story, and it is now fixed. Under a scan charge — the predictor re-evaluating φ from scratch at every step — the same five rules read 0.119–0.361 and look strong. Under the incremental charge they read 0.472–0.789. The definition of κ is a supremum over predictors with $O(\log n)$ bits, and an optimal predictor maintains the statistic, so the incremental charge is the one the definition implies. Both numbers are correct statements about different predictors; only the second is κ.

What the table rests on. The walk grows like n^2, so most tapes halt or repeat before they output: at $n = 512$ the cells rest on 373 to 1 835 meeting tapes. A cell is good to about ± 0.07, and no ordering of two rules that differ by less than 0.15 should be read off the table. Every entry is a lower bound: the predictor is a ridge regression on 41 bits, and a stronger device at the same budget raises it — for `runs`, the CES device at 3.4–4.1 reads per cell reaches 1. That device is not only a predictor: it is exact on the meeting tapes (Section 3), which is why `runs` and `pattern` are

rule	κ_ρ at $c = 16$, $n = 512 \to 4096$	trend
`trip`	$0.789 \to 0.472$	falling
`peak`	$0.785 \to 0.614$	falling
`pattern`	$0.693 \to 0.502$	falling
`dense2`	$0.632 \to 0.509$	falling
`runs`	$0.569 \to 0.520$	falling

Table 5: κ_ρ under the incremental charge: the predictor is the ridge regression on the state features of Appendix A.2 (constant predictor always in the table), it maintains the rule's statistic and pays one to four reads per step, not m. At $c = 16$ and $n = 512 \to 4096$ all five rules fall in n; none reaches 1 with *this* predictor, and at $c = 128$ the largest is 0.999 at $n = 512$ (0.927 at $n \geq 1024$). These are lower bounds from one family of predictors, and for two of the five the bound is not tight: the CES device of Section 3 reaches $R^2 \geq 0.99$ at 3.4–4.1 reads per cell on `runs` and `pattern`, which gives $\kappa = 1$ for both and is why the class table places them at full linearity rather than in the superlinear band. The eight rules not listed are being re-measured under this charge: the scan charge prices the same predictor m reads per step instead of one to four, which moves the values by a factor of the window length; those numbers are not comparable and are not printed.

fully linear and not merely predictable. A κ of 1 therefore signals a cheap *machine*, not a separate class.

Conjecture 5 (The threshold)**.** *There is a constant $\kappa^\star < 1$, independent of the rule and of n, such that every weak-superlinear rule satisfies $\kappa_\rho \leq \kappa^\star$. Equivalently, a rule whose measured κ exceeds $\kappa^\star$ has a cheap exact machine: a predictor exists that drives its cost to the machine's, $\kappa \to 1$. The boundary between the weak-superlinear band and full linearity is a single number, which makes the classification decidable by measuring κ and comparing it with $\kappa^\star$.*

The measured values run from 0.50 to 0.80 across the five rules re-measured under the incremental charge. The device that used to supply the upper witness — `runs` at 3.4–4.1 reads per cell — turns out to be exact on the meeting tapes, so it places `runs` in the fully-linear class rather than above $\kappa^\star$. What the data therefore fix is only the lower end: $\kappa^\star \geq 0.80$, above the largest weak value. The upper end is open, and with the second reading measured empty the conjecture's two readings — a threshold between weak-superlinear and full linearity, or a threshold between superlinear and the existence of a cheap exact machine — are no longer distinguished by any rule in the catalogue.

5.1 Two arguments that cannot work

Theorem 3 (Information ceiling)**.** *The device's answer at a query is one of eight actions, so a query carries at most $\log_2 8 = 3$ bits. One read yields at most $\log_2 3 = 1.585$ bits on the three-valued (sum) alphabet, and 2 bits on the four-state cell; the two give ceilings of $3/1.585 = 1.893$ and $3/2 = 1.5$ reads per query. No information argument on either alphabet can force more than 1.893 reads per query, whatever φ is; the three-valued count gives the ceiling 1.893 and the four-state count 1.5; the larger ceiling is the stronger statement, so the three-valued count is the more powerful of the two, and neither comes close to the costs the rules actually incur.*

The ceiling is not a statement about a weak argument; it is a statement about the model. A machine that answers a query with one of eight actions cannot be *proved* to read more than 1.893 cells per query by any counting argument — the argument's ceiling, not the machine's requirement — because on the three-valued alphabet the answer carries at most $\log_2 3$ bits per cell read.

Definition 2 (Certificate)**.** *A* certificate *for a query at a given state is a set of cells whose values determine the query's answer at that state. Write* $\mathrm{cert}_\Delta(\tau)$ *for the size of the smallest certificate needed by a device that answers every query exactly.*

Theorem 4 (Certificate ceiling)**.** reads $\geq \mathrm{cert}_\Delta(\tau)$*, and* $\mathrm{cert}_\Delta \leq n$ *for every rule, so no certificate argument can prove more than* n *reads. Certificates also do not compose: a cell whose flip changes any earlier answer must be read at that query, so the per-query certificates collapse into a single union. On the constant tape at* $n = 1024$*,* $\sum_t |J_t| = 1\,285\,703$ *while* $|\bigcup_t J_t| = 1\,020$*, a factor* 1260*.*

The second ceiling is the sharper constraint. A lower bound that proceeds by counting, for each query, the cells that would have to be read, silently assumes those sets are disjoint or nearly so; they are not, and the union is smaller than the sum by three orders of magnitude. Any proof of super-quadratic cost has to use something that is neither a counting argument nor a certificate.

5.2 The gap between the output and the action

A lower bound is usually proved for the rule's output φ; what the walk consumes is its action δ. The two differ, and the difference is measurable. Call $\bar{p}$ the fraction of length-weighted queries at which flipping a single answer moves δ. Exact single-answer replay on the real walk gives

$$\bar{p} = 0.104,\ 0.078,\ 0.0153,\ 0.0072,\ 0.0071,\ 0.0042 \qquad (n = 128, 256, 512, 1024, 2048, 4096),$$

a slope of -0.99, and $\bar{p}_{\mathrm{trim}} = 0$ exactly: at a trim the answer is not merely irrelevant, it is unused. A bound proved for φ therefore transfers to the walk only after multiplication by $\bar{p}$: the walk pays for the $\bar{p}$-fraction of queries whose answer can change its action, and for no others.

One bound escapes this. The resolving-pointer bound is stated directly for a device that takes the rule's action at every step, so it is already a statement about δ; it gives $\Omega(n^2)$, and it is the one super-linear lower bound in the programme that does not shrink when the gap is accounted for. The class-restricted bounds of the literature, restated for δ, carry slopes 1.90 and 1.91 rather than 3.

6 Where this sits in the classical picture

Simulating several heads with one. This is an old problem. A multi-tape or multi-head machine can be simulated by a single-head one in quadratic time [Hartmanis and Stearns, 1965], by a two-tape machine in $O(n \log n)$ [Hennie and Stearns, 1966], and Hennie's crossing-sequence argument gives an $\Omega(n^2)$ lower bound for a single tape that recognises the palindromes [Hennie, 1965]. The fastest possible simulation is therefore known to lie between a nonlinear lower bound and the quadratic upper bound, and the gap has stayed open [Kutrib, 2010]. Our measurements sit inside that gap: with $O(\log n)$ bits the canonical rule costs $147.7\,n$ reads at $n = 1024$ (the `pcr_lib.Runner` device), and the five level-count rules of Section 4 cost at most $1.33\,n^2$ (measured 1.12).

Multi-head automata and pebbles. Two-way finite automata are no stronger than one-way ones [Rabin and Scott, 1959], but adding heads does add power: $k + 1$ heads are strictly stronger than k [Rosenberg, 1966], and the hierarchy survives even when the heads carry no state [Kutrib and Malcher, 2009]. A machine that can tell when two of its heads coincide is a sensing-head automaton [Petersen, 1995], and a machine that may drop markers on the tape is a pebble automaton [Blum and Hewitt, 1967]. Our machine sits between the two: it has two heads, it can detect coincidence,

and it can trim, but a trim is not a pebble. A pebble is placed where the machine chooses and can be lifted; our two boundaries only move inward, the place where they stop is set by the first exit of the walk rather than chosen, and a trim destroys the cells behind it for good. That is why the questions asked here — how many reads a given workspace buys, and how much of the output a given amount of memory determines — do not reduce to the known hierarchy results.

Reproducibility

Every number in this paper is the output of a script. The scripts, the machines and the verification harnesses are collected at `https://github.com/peuaza/triangular-window-walk`, which also carries the simulations of the earlier release; the release matching this manuscript is tagged. The measurements of Section 4 were made at $n = 4096$, 8192 and 16384; the verification figures quoted in the tables are the counts of comparisons made at every query, not sample sizes. The appendices collect the protocols, the verification gates, the technical lemmas, and the status of each claim.

The numbers of Section 3 come from five probes, each a directory in the same repository with its own report: `probe_ces_exact` (the CES device, its exactness on the meeting tapes, and the linear costs of `pattern` and `runs`), `probe_en_linear` (the same question put to the other fifteen rules, and the three failure mechanisms), `probe_k300` (the band family at 300 meeting tapes per cell), `probe_g_witness` (the nine-tape witness panel and the g exponents), `probe_peak_fix` (`peak`'s exactness and its G/e), `probe_semilinear` (the semi-linear construction), `probe_cx_down` (the anchor form $A = s/31$ and the $n^3/\log n$ law of `CX`) and `probe_audit` (the check that each number above traces to a script) and `probe_three_order2` (the obstruction to a three-order rule). Each report names, per table, the JSON that carries the number and the machine and charge rule that produced it.

Use of AI assistants

The computational work reported here — the machines, the verification harnesses, the measurement drivers and the checks that produced every number in the tables — was written with the assistance of an AI coding agent, under the author's direction. The prose was drafted with the same assistance and revised by the author. The author set the questions, adjudicated every mathematical claim, rejected several intermediate conclusions that the measurements later contradicted, and verified each reported number against a script. No AI system is an author of this work, and no reported number was generated by a language model: every one of them is the output of the code collected with the paper.

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A The model, the predictor, and the proofs

This appendix does three things the main text does not have room for. It fixes the model formally (A1), including which of the two readings of φ is in force at each point — a distinction the text uses throughout without naming. It defines the predictor and the quantity κ that the classification is stated in (A2). It then proves, or cites the verification of, the structural facts the paper relies on (A4), and records the protocols (A5), the gates (A6) and the status of each claim (A7).

A.1 The model

Definition 3 (Rule)**.** *A* rule *is a pair* $\rho = (\varphi, f)$ *with*

$$\varphi : W \to A, \qquad W = \bigcup_{m \geq 1} \Gamma^m, \qquad A \textit{ finite},$$

$$f : A \times \mathbb{N} \times \mathbb{N}^4 \longrightarrow \{\ell{+}1,\ \ell{-}1,\ r{+}1,\ r{-}1,\ \textit{left trim},\ \textit{right trim},\ \textit{double trim},\ \textit{halt}\},$$

evaluated at $\big(\varphi(\tau[\ell..r]),\, n,\, \ell,\, r,\, L,\, R\big)$*. Two conditions hold by definition and are used silently below:* φ *depends on the current window only, never on the history, so* (φ, f) *makes the state* (L, R, ℓ, r) *Markov; and every convention by which a rule resolves an inadmissible move, a trim or a halt is part of* f*. No computability, complexity or measurability condition is placed on* φ*: it is the* specification *of the rule, not an object the rule computes.*

Definition 4 (Run)**.** *Fix* n *and* $\tau \in \Gamma^n$*. The state is* (L, R, ℓ, r) *with* $0 \leq L \leq \ell \leq r \leq R \leq n-1$*, initially* $(0, n-1, 0, n-1)$*. Given a state, let* $a = f(\varphi(\tau[\ell..r]), n, \ell, r, L, R)$*.*

1. *If* $\ell = r$*, the run halts and outputs* $\delta = \ell/(n-1)$*.*
2. *If* a *is a move taking a header to* p*, it is admissible iff* $L \leq p \leq R$*. If admissible and* p *meets or passes the other header, the run halts and outputs* $\delta = p/(n-1)$*; if inadmissible the run aborts with no output.*
3. *If* a *is a trim, it is admissible iff the new region is non-empty; then* $(L, R) \leftarrow (L', R')$ *and* $(\ell, r) \leftarrow (L', R')$*.*
4. *If* a *is* halt*, the run halts with no output.*
5. *If the new state has occurred earlier in the run, the run halts with no output.*

Definition 5 (Output function)**.** $\delta_{\rho,n} : \Gamma^n \to D_n \cup \{\perp\}$*, where* $D_n = \{i/(n-1) : 0 \leq i \leq n-1\}$ *and* $\perp$ *denotes "no output", is the map defined by Definition 4. It is total: the state space is finite, so every run halts by clause 5.*

Remark 1 (The repeat test costs no space)*.* Clause 5 is a property of the orbit, not a resource of the device: f and the geometry determine the next state, so a device can implement the test by Brent's algorithm in $O(1)$ extra work bits. It costs reads and steps, never space. The clause therefore does not smuggle an $\Omega(n^2)$-bit table into the model — but a device that merely simulates must pay for cycle detection, and the read counts of this paper include that payment.

A.1.1 Two readings of φ, and which is in force where

Definition 6 (Oracle semantics, (O))**.** *φ is an oracle: evaluating $\varphi(w)$ costs 0 reads and 0 steps, for every window w.*

Proposition 4 (The oracle reading is vacuous)**.** *Under (O), every rule ρ costs zero reads, so every rule is trivially fully linear and none is superlinear.*

Proof. The device stores (L, R, ℓ, r) — four integers below n, so $4\lceil \log_2 n \rceil$ bits — queries the oracle for $\varphi(\tau[\ell..r])$, looks f up in its finite control, and updates. It never senses a cell, so reads = 0. Clauses 1–5 of Definition 4 are decided from the stored state and the oracle's answer. □

Definition 7 (Computed semantics, (C))**.** *The device has no oracle. On input τ it must write $\delta_{\rho,n}(\tau)$ on its output tape, sensing only cells of τ. φ enters only through $\delta_{\rho,n}$; the device may compute φ however it likes, precompute it, cache it, or bypass it entirely.*

Definition 8 (φ-state; incremental computability)**.** *A* φ-state *for ρ on a window w is a value $\sigma \in \Sigma$ together with an evaluation map* $\mathrm{ev} : \Sigma \times \mathbb{N} \to A$ *such that* $\mathrm{ev}(\sigma, |w|) = \varphi(w)$*. A device* carries φ *if its work tape holds a φ-state and* ev *costs $O(1)$ steps and 0 reads. φ is* incrementally computable *if there is* $\mathrm{upd} : \Sigma \times \Gamma \to \Sigma$*, costing $O(1)$ steps and 0 reads, with* $\mathrm{upd}(\sigma_w, c) = \sigma_{w\setminus c}$ *for every window w and every cell c at either end of w.*

Remark 2 (Which reading is in force where)*.* The three readings occur in different parts of the paper, and the classification is only meaningful under (C).

location	reading	what it establishes
Definition 3	(O)	the *semantics* of the rule: f is handed $\varphi(\tau[\ell..r])$. Not a cost statement.
the machines of Section 4	(C)	read counts of devices that compute φ themselves; every number in the tables is of this kind.
Theorem 1, Theorem 2	(C) with a carried φ-state	the only places where the assumption is written down; strictly stronger than Definition 3.
the ceiling of Section 5.1	neither	a statistical estimator with a read budget, not required to be exact; not a device in the sense of Definition 3.

The consequence of Proposition 4 is worth stating plainly: a paper that counted reads under (O) would classify nothing, since every rule would cost zero. Every read count below is therefore a statement under (C), and the two theorems that mention a carried φ-state are the only places where that assumption is used at full strength.

A.2 The predictor, and κ

The classification uses κ, and κ is a supremum, so it needs both a definition and a statement of what a measurement of it can and cannot establish.

Definition 9 (Predictor)**.** *Fix ρ, n and a budget $B = B(n)$. A* predictor *is a device that, on input τ, reads at most $B(n)$ cells of τ — adaptively, in any order — performs arbitrary computation on what it has read, and outputs one value in D_n. It is not required to be correct; it is scored. The tapes that concern it are the* meeting *tapes, those on which $\delta_{\rho,n}$ is defined: a tape with no output has no δ to predict and is not counted.*

Definition 10 (Score, and κ)**.** *For a predictor P let $R^2_n(P)$ be the coefficient of determination of P's output against $\delta_{\rho,n}$, computed over the meeting tapes with cross-validation and with the constant predictor in the comparison. Define*

$$\kappa_\rho(c) = \lim_{n\to\infty} \sup_{P\,:\,B=cn} R^2_n(P), \qquad \kappa_\rho = \sup_{c\geq 1} \kappa_\rho(c).$$

The classes are read in this order: fully linear *if $g_\rho = O(n)$; otherwise* semi-linear *if $e_\rho = O(n)$; otherwise* superlinear*. The superlinear class is not split further: κ_ρ measures how close a rule comes to a cheap exact machine, no member of the catalogue reaches $\kappa_\rho = 1$, and Conjecture 2 says none can.*

Remark 3 (What a measurement of κ can establish)*.* κ is a supremum over predictors and a limit in n, so a measurement is a *lower* bound: it exhibits one predictor and reports what it achieved. Two consequences follow.

1. $\kappa = 1$ *can* be established, by exhibiting a device that reads at most cn cells and reproduces δ on every meeting tape.
2. $\kappa = 0$ *cannot* be established by measurement. The strongest statement a table can carry is that every predictor tried failed to beat the constant predictor.

Proposition 5 (Linear mean forces perfect prediction)**.** *If $e_\rho = O(n)$ then $\kappa_\rho = 1$.*

Proof. Let M be a device with mean cost $e \leq Cn$ over the meeting tapes. Cap it at cn reads and let it guess on the tapes where the cap stops it. By Markov's inequality the capped fraction is at most C/c, and since $\delta \in [0,1]$ the squared residual on those tapes is at most one, so

$$R^2_n(c) \;\geq\; 1 - \frac{C}{c\,\mathrm{Var}(\delta)}$$

for every n and every c. Taking $n \to \infty$ and then $c \to \infty$ gives $\kappa_\rho = 1$. □

The contrapositive is what the classification uses: $\kappa_\rho < 1$ implies $e_\rho = \omega(n)$, so every rule with $\kappa_\rho < 1$ is superlinear. The converse fails — a rule with superlinear mean cost can still be asymptotically predictable — and that failure is exactly the boundary at $\kappa = 1$.

Remark 4 (R^2 is blind to one-cell errors)*.* δ takes values on a grid of spacing $1/(n-1)$. A predictor that is off by exactly one grid cell on a constant fraction of tapes still has $R^2 \to 1$, because the rescaled residual is $(n-1)\,|\hat{\delta} - \delta| = 1$ and its square is $1/n^2$ against a total variance of order one. A rising R^2 therefore does not show that the misses are shrinking. The quantity to report is the residual *in grid units*: the fractions with $(n-1)|\hat{\delta} - \delta|$ equal to 0, to 1 and to at least 2, and how those fractions move with n.

A.3 The catalogue of rules

Every rule the main text names appears in Table 6, one row each, with its φ written on a single line precise enough to reimplement. Write $m = r - \ell + 1$, $k = \max(1, \lfloor m/2 \rfloor)$ and $a_i = 2\tau_i \in \{0,1,2\}$; let s be the label tape and G its prefix walk, $G_0 = 0$, $G_{j+1} = G_j + s_j$. Except where a row states otherwise, $\varphi = (b_2, b_1)$ with the concavity test $b_2 = b_2^{\mathrm{conc}} = [\,k\sum_{i=0}^{m-1} a_{\ell+i} > m\sum_{i=0}^{k-1} a_{\ell+i}\,]$ as its second coordinate, so each cell of the table records b_1 alone. The rows marked four-valued read the label tape s; the three-valued rows read only $\sigma = a$; by the embedding lemma every σ-rule

is bit-identical in the four-valued model, so the two columns are a statement about what the rule needs, not about which engine ran it.

The action convention is the canonical four-move table — $\ell \mapsto \ell - 1$ on $(0,0)$, $r \mapsto r + 1$ on $(0,1)$, $r \mapsto r - 1$ on $(1,0)$, $\ell \mapsto \ell + 1$ on $(1,1)$, the headers moving inward exactly when the two bits agree — unless the row says otherwise, and three rows do. `hashmod` uses a two-action map that ignores b_2: $b_1 = 0$ moves $r \mapsto r - 1$ and $b_1 = 1$ takes a left trim. `PC` and `CX` use a three-clause convention on (m, W) with $W = R - L + 1$: move $\ell \mapsto \ell + 1$ when $m \le 2$, move inward on b_1 when $W - m < k(W)$ with $k(W) = \max(1, \lfloor W/64 \rfloor)$, and otherwise trims on the side fixed by the parity of $\ell - L$. Two exit-trim conventions occur in the record, and they differ whenever the window does not touch the boundary it leaves through. The canonical engines trim the side *opposite* to the one the move would have left; the sign-blind and the four-valued engines trim the *same* side. Only the canonical row carries a dagger below and uses the first; every other four-move row uses the second.

Three of these rows — `BAL`, `KCHG` and `LAS` — have no prose definition anywhere in the record: the manuscript, the measurement notes and the appendix all name them and none defines them. Their φ above is taken from the implementation rather than from a stated definition, so a reader who wants to check them should read the code, and should expect the definition it finds to be the one that produced the numbers rather than the one a gloss would suggest. `LAS` is the sharpest case: as implemented it coincides with `runs`, which is why its separate classification rests on measurement alone.

A.4 Structural facts, and what of them is proved

Each item below is labelled *proved* or *verified*. The distinction is not decorative: three of the six are verified on exhaustive small cases and are not theorems, and saying so is cheaper than having a referee say it.

Parity of the boundary (proved). Write C for the number of cells of the window exceeding its mean and $D = 2C - m$ for the signed surplus. Since $2C \equiv m \pmod 2$, $D \equiv m \pmod 2$, so the boundary $D = 0$ — the only state in which a scan can change an epoch's outcome — is reachable only at even m. Verified on all 27 984 epochs of the cube for $n = 4$–8: $b_2 = 0$ forces a double trim without exception, and every epoch whose outcome differs from the b_1-side trim visits $D = 0$.

Drift (proved for even n). For the ensemble of Definition 3, $\mathbb{E}[4N_2 - m] = 0$ at even n and $+2$ at odd n, where N_2 counts the cells equal to 2. Hence `dense2`'s bit has no drift at even n, flips in 35 per cent of epochs, and flips the trim side with it. Measured consequence: enlarging the endgame buffer from 24 to 320 cells leaves the error rate at 14 per cent.

Level counts for the extremum rules (verified). For a ± 1 walk, the statement "the endpoint r is the strict maximum of the window" is equivalent to "the level G_{r+1} is never visited in $[\ell, r]$". The equivalence turns `AMAX` and `AMIN` into single-level counts, which is what makes them reachable by the same band+frontier machine as `NBLK`. Verified on 132 036 windows with no mismatch. This is a verified equivalence, not a proved one; the converse direction is the one that could fail on a longer window, and it was checked only on the windows the walk visits.

The rank of the convolution kernel (proved). For $h(k) = 3^{k^2}$ and modulus q,

$$H_m[d,k] = h(k+d) = 3^{(k+d)^2} = 3^{d^2} \cdot 9^{dk} \cdot 3^{k^2},$$

rule	tape	φ	class
`canonical`†	3-valued	$\big[\,\widehat{\mathrm{corr}}_1(a_\ell,\dots,a_r) > 0\,\big]$, with $[a_\ell > \frac{1}{2}]$ at m=1, $[a_r > a_\ell]$ at m=2, and 0 when either lagged half is constant	FL
`monotone family`	any	none: the family is defined by f, not by φ	FL
`NBLK` (statistic)	4-valued	$\mathrm{NBLK}(w) = \#\{\, i \in (\ell, r+1] \;:\; G_i = G_\ell \,\}$ — the returns of the label walk to its starting level	—
`NBLK=1`	4-valued	$\big[\mathrm{NBLK}(w) = 1\big]$	SU
`NBLK>2`	4-valued	$\big[\mathrm{NBLK}(w) > 2\big]$, count ≥ 3	SU
`NBLK>3`	4-valued	$\big[\mathrm{NBLK}(w) > 3\big]$, count ≥ 4	SU
`AMAX`	4-valued	$\big[G_{r+1} > \max_{\ell \leq i \leq r} G_i\big]$, the window's border attains a strict maximum	SU
`AMIN`	4-valued	$\big[G_{r+1} < \min_{\ell \leq i \leq r} G_i\big]$	SU
`MAXP`	4-valued	$\big[(\max_{\ell \leq i \leq r+1} G_i - G_\ell)^2 > m\big]$	SU
`KCHG`	4-valued	$c = \#\{p \in [\ell+1, r] : s_{p-1}s_p < 0\}$, p_c = the $\lceil c/2 \rceil$-th such p; $\big[2(p_c - \ell) > m\big]$, and 0 if $c = 0$	SU
`LAS`	4-valued	$\mathrm{LAS}(w) = 1 + \#\{\text{sign changes in } (s_i)_{i \in w,\, s_i \neq 0}\}$, (0 if none); $\big[4\,\mathrm{LAS} > m\big]$	SU
`BAL>m/2`	4-valued	$\mathrm{BAL}(w) = \max\{j - \ell : \ell < j \leq r+1,\ G_j = G_\ell\}$, (0 if none); $\big[2\,\mathrm{BAL} > m\big]$	SU
`BAL>m/4`	4-valued	same BAL; $\big[4\,\mathrm{BAL} > m\big]$	SU
`hashmod`	3-valued	$\big[H(w) \bmod m = 0\big]$, $H(w) = \big(\sum_{j=0}^{m-1} 2^j a_{\ell+j}\big) \bmod p$, $p = 10^9 + 7$	SU
`dense2`	3-valued	$\big[4\,\#\{i \in [\ell, r] : a_i = 2\} > m\big]$	SU
`runs`	3-valued	$\big[2\,\#\{i \in [\ell, r-1] : a_i \neq a_{i+1}\} > m\big]$	SU
`pal`	3-valued	$\big[a_\ell a_{\ell+1} \cdots a_r = a_r \cdots a_{\ell+1} a_\ell\big]$	FL
`pattern`	3-valued	$\big[\exists j \in [\ell, r-2] : (a_j, a_{j+1}, a_{j+2}) = (2,1,1)\big]$, and 0 if $m < 3$	FL
`longrun`	3-valued	$\big[4\,\Lambda_2(w) > m\big]$, $\Lambda_2(w) = \max\{t : \exists j,\ a_j = \cdots = a_{j+t-1} = 2\}$	SU
`peak`	3-valued	$\big[6\,\#\{i \in [\ell+1, r-1] : a_{i-1} < a_i > a_{i+1}\} > m\big]$	SU
`trip`	3-valued	$\big[6\,\#\{i \in [\ell, r-2] : a_i = a_{i+1} = a_{i+2}\} > m\big]$	SU
`oddrun`	3-valued	$\big[2\,\#\{\text{maximal constant runs of } w \text{ of odd length}\} > m\big]$	SU
`ballast`	3-valued	the monotone collapse of Section 3 with the one-shot defect defined there	SL
`PC`	4-valued	$p_0 = 0$, $p_{t+1} = (3p_t + 1 + a_{\ell+p_t}) \bmod m$, $\big[2p_m \geq m\big]$	—
`CX`	4-valued	$q = 2^{31} - 1$, $h(k) = 3^{k^2+1} \bmod q$, $B(w) = \sum_{i=\ell}^{r} a_i\, h(i-\ell) \bmod q$, $\big[2B < q\big]$	—

Table 6: The catalogue. Each row is one rule with the tape it reads, its φ in one line, and the class it is placed in: FL fully linear ($g_\rho = O(n)$), SL semi-linear ($e_\rho = O(n)$ with $g_\rho = \omega(n)$), SU superlinear ($e_\rho = \omega(n)$), — unclassified. The superlinear rows are not subdivided here: their measured κ_ρ is in Table 5, and every one of them is strictly between 0 and 1. A dagger marks the one row that uses the opposite-side exit trim; row 1 of the table is the only such row.

so $H_m = 3\,\mathrm{diag}(3^{d^2})\,[9^{dk}]\,\mathrm{diag}(3^{k^2})$. The middle factor is a Vandermonde matrix on the nodes 9^d, $d = 0, \ldots, m-1$, which are pairwise distinct whenever $\mathrm{ord}_q(9) > m-1$. With $q = 2^{31}-1$, the order is 357 913 941 — computed, not assumed: q is prime, the order divides $q-1$, and it exceeds the largest window the tables use by a factor of 21 845 (`probe_vandermonde/vand_check.py`) — so the nodes are pairwise distinct for every m in the range studied and $\mathrm{rank}\, H_m = m$ exactly over $\mathbb{F}_q$. The two diagonal factors are powers of 3, nonzero in $\mathbb{F}_q$ and hence invertible, so the rank is that of the Vandermonde factor. This is the only rank computation the paper relies on.

The witness family for the cubic cost (verified). The family is 1^m — the constant tape — with nine cells changed. The machine's two encodings cover up to eight defects and it uses nine, which its accumulator still absorbs; the cost per test is flat in the window length for every fixed count that was checked, up to sixteen. On this family the Brent-style machine costs $n^3/685$ at $n = 4096$ with local exponent 2.90, and $C(m)/m$ is flat in m (per-window exponent 1.07 on the artificial family, 0.95 on the real tape). The dependence on the defect *count* rather than density is what makes the exponent 3 rather than 2.5; with a fixed defect density the same machine gives 2.5 to 2.85. This is a measured witness, not a lower bound: no argument in this paper shows that no better machine exists.

Why one number suffices (proved). If b_1 is constant over the epoch's range and that constancy is certified by a tape-global maximum, the epoch's effect is a single trim whose side is a function of (b_2, b_1) evaluated once, so the strategy costs $O(n\rho)$, where ρ is the longest run of 2s — this is `longrun`. If the certificate is instead a last-passage query on a ± 1 walk, its exact structure needs $\Theta(n)$ words and the strategy does not linearise. The dichotomy is the reason the certificate's *shape*, not its size, decides the class.

Step counts, and why there is no clock The body states no rule-independent clock, because the measurement contradicts it. On one ensemble, with every run allowed to finish, the mean step count at $n = 4096$ divided by $n^2/128$ is:

The ratio ranges from 0.385 for `NBLK>3` to 15.58 for `hashmod`, a factor of forty, and the growth exponent between $n = 1024$ and 4096 ranges from 1.70 to 2.20. These are the step counts of the machines of Section 4, regenerated from their traces; an earlier draft printed a wider span whose low end was not in the data. A clock would have to be a property of the model; this is a property of the rule.

The collapse (verified, not proved). The collapse (Proposition 1) was checked on 320 880 queries with no error, and the body states it as a proposition for that reason; the general-n statement is Conjecture 1.

The general bound, and why it is not sharp. In cell reads the trivial bound is $O(n^4)$: the walk makes $O(n^2)$ steps and a step that re-evaluates a window statistic costs up to $O(n)$ reads. Under the epoch structure the bound is $O(n^3)$, and the five level-count rules of Section 4 meet the next rung down at n^2 with the band+frontier machine. The n^3 rung is occupied by `PC` on the witness family above. What is missing is not a better machine but a lower bound: Section 5.1 records the two ceilings that block the standard arguments.

A.5 Protocols and gates

Every machine is written read-exact and gated before any timing, at zero tolerance: the walk against the literal definition on the full cube $\{0,1,2\}^n$ for $n \le 10$ (88 560 tapes) and on random tapes at $n = 11$–96, accepted only at zero mismatches in δ *and* in the step count; the b_1 table against an independent implementation on 2 192 103 windows; and the registers after every trim against a prefix-sum oracle. Three bugs in our own machines were caught this way after their numbers had been computed once, and the numbers were recomputed.

Two protocol choices are recorded because each changed a number. Exponents are fitted on the largest three sizes only: fitting from $n = 64$ produced two exponents we later withdrew (1.696 for `peak` and 1.826 for `trip`; both are 2.04–2.12 on the large sizes), because the $2n$ initialisation dominates at small n. And the worst case is measured on a constructed family rather than by hill climbing: on `NBLK>2` a climb found $4.3\,n^2$ where the family found $48.9\,n^2$ at the same size, a factor of eleven.

A.6 Verification gates

Every machine used for a number in this paper passed a gate of the following form before any timing. The counts are totals over the paper's strategies.

what is checked	how	count
the walk against the literal definition	full cube $n \le 10$	88 560 tapes
the same, four-valued model	cube + support T_n	130 500 walks
the machine's registers after every trim	prefix-sum oracle	every scan step
b_1 on all windows of the cube	independent implementation	2 192 103 windows
the endgame at $W = n$ reduces to the walk	cube $n \le 10$	147 555 walks
the constructed worst-case family is realisable	by construction	all members

Table 7: Gates. A number is reported only if its machine passes. Three real bugs in our own machines were caught this way after the numbers had already been computed once, and the numbers were recomputed.

A.7 Status of each claim

A.8 The ensemble

A tape of length n is drawn by choosing, independently for each of two columns, a uniform h-subset of $\{1,\dots,n\}$ with $h = \lceil n/2 \rceil$; a cell's value is the sum of its two bits, so values lie in $\{0,1,2\}$ with probabilities $(1/4, 1/2, 1/4)$ up to the dependency induced by the subsets. All averages are over this ensemble. Strategies read only $\sigma = b_1 + b_2$ except where the text says the label $s = b_1 - b_2$ is used.

claim	status	where
parity of the $D = 0$ boundary	proved	Appendix A.4
`dense2` has no drift at even n	proved	Appendix A.4
$g_\rho = O(n^4)$ in cell reads, $O(n^3)$ under the epoch structure	proved	Appendix A.4
$g_\rho = \Theta(n^3)$ for machines that recount per test	proved	Appendix A.4
global-max certificate $\Rightarrow O(n\rho)$	proved	Appendix A.4
`longrun`: e/n rises $8.53 \to 9.39$ over $n = 512 \to 16384$, $\alpha = 1.024$	measured	Section 3
`runs` costs $2.77\,n$ on the CES device, exponent 1.00	measured	Section 4
$\kappa \in [0.472, 0.925]$ for thirteen rules at $c = 16$	measured	Section 5
$\kappa_{\texttt{hashmod}} \approx 0$	measured	Section 5
no strategy has $g_\rho = \omega(n^3)$	**open**	—
every rule outside the fully linear class has $g_\rho = \omega(n)$	**open**	—
"every non-linear rule has $g_\rho = \Omega(n^2)$"	**false**	the constructed $n^{3/2}$ rung ($g = \Theta(n^{1.5})$) and `ballast` ($g = \Theta(n^2)$)
some rule costs $n^3/685$ on the witness family	measured, upper bound	Section 4
"a window statistic can be evaluated in $O(1)$ reads"	**false in general**	Appendix A.4

Table 8: What is proved, what is measured, and what is open. "Measured" means the protocol of Appendix A.5 ran and the gate of Appendix A.6 passed; it does not mean a lower bound was proved.